

SECTION B

6 (short)

- (a) Describe what is meant by 2's-complement representation of a binary number and explain its significance in digital information processing. [2]
- (b) Design a circuit that converts a 3-bit binary 2's-complement number X to its negative $-X$, represented as a binary number in 2's-complement form. The circuit should be composed only of NAND gates and the smallest possible number of such gates should be used. [8]